WASP-12b

R-0550

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-12_180210` · created 2026-07-22T08:45:28+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458159.77513 +/- 0.00046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.09 +/- 0.026
Transit depth (Rp/Rs)^2	0.81 +/- 0.47 [%]
Orbital Inclination (inc)	81.81 +/- 2.79
Airmass coefficient 1 (a1)	1.313 +/- 0.023
Airmass coefficient 2 (a2)	-0.1948 +/- 0.0049
Scatter in the residuals of the lightcurve fit is	1.71 %
Best Comparison Star	None
Optimal Aperture	5.5
Optimal Annulus	7.63
Transit Duration (day)	0.1165 +/- 0.0081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.09 ± 0.026 vs 0.1178 (deviation 0.63σ)
Duration fit vs published	0.1165 d vs 0.125 d (deviation 0.62σ)
Mid-time O–C	-1.2 min
Depth SNR	2.0
Residual scatter	1.71 %

Light curve

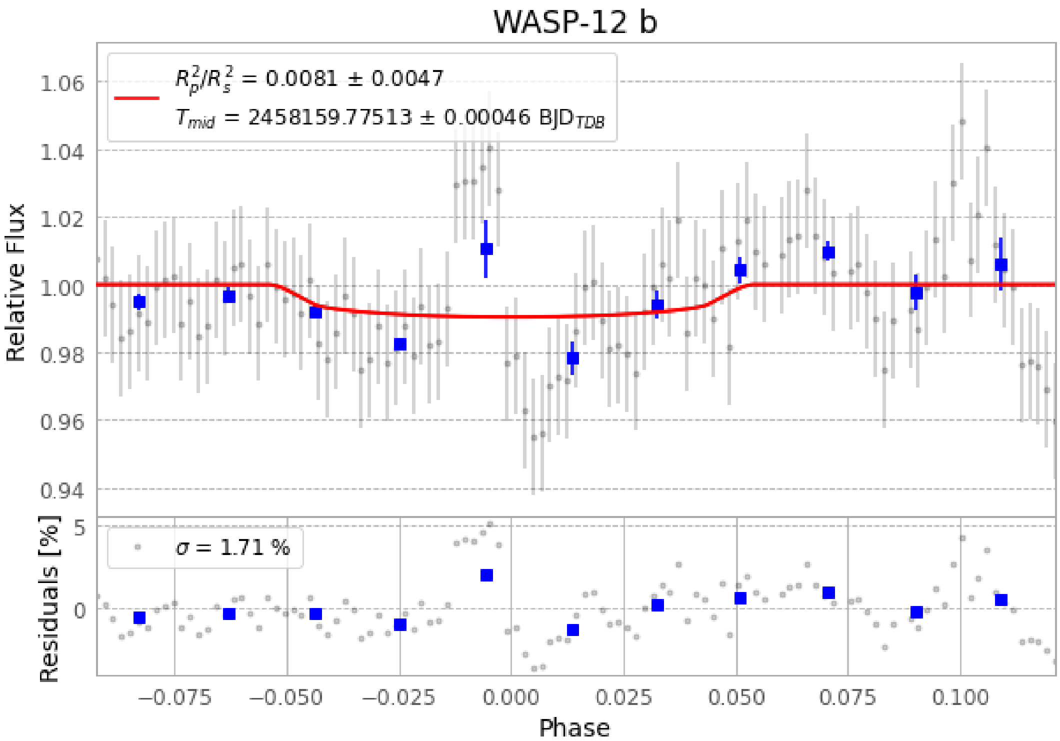

Figure 1 — FinalLightCurve_WASP-12 b_2018-02-09.png (SHA-256 1d52d31ed4c84403…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [347, 270], 1 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 33bdf29bf322cd13…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494`

Data provenance

114 FITS frames (75 MB), `WASP-12180210040314.FITS` → `WASP-12180210094212.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-180210.log (26ba5f59e1e4…), FinalLightCurve_WASP-12 b_2018-02-09.png (1d52d31ed4c8…), FinalLightCurve_WASP-12 b_2018-02-09.csv (e0fa5b32dc85…)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-22 08:45Z.